# Graph-Aware Planning-Assisted Learning for Distributed Personalized Manufacturing

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**Abstract**

In a synthetic twenty-site manufacturing network, a graph convolutional network (GCN) corrects an eight-step planner's (H8) candidate costs using paired Monte Carlo targets. A five-block, 120-world comparison separates operational, learning, and graph effects. The full controller reduces mean cost by 4.845% relative to the MDL2 heuristic, with 94.567 fewer patient losses and 110.942 more completions per world. Relative to H8, cost falls by 2.136% overall and is lower in every block mean, but two conditions have additional patient losses. The matched identity-adjacency comparison yields a smaller 0.407% reduction, with a descriptive 95% interval of $[-0.087, 1.041]$% and block disagreement. A separate three-block, 24-world study removes global H8 from both models and matches their bounded messages, parameter counts, and fitting budgets. Here the GCN costs 4.843% less than the non-graph model, with lower cost in all three block means, but incurs 16.667 additional patient losses and 12.375 fewer completions per world. The primary study supports an operational benefit and a learning-related cost increment; the message study supplies a cost-only architecture signal. Neither establishes service-preserving independent GCN gains or explains the different graph effects across these distinct interfaces. Findings remain synthetic development evidence, not clinical validation.

*Keywords:* Planning-assisted learning, Monte Carlo policy improvement, Graph convolutional networks, Patient-service accounting, Personalized manufacturing, Synthetic simulation

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## 1. Introduction

Personalized manufacturing requires coordination between patient-specific demand and distributed production resources. A specimen and its resulting product cannot be reassigned to a different patient. Moving a specimen may make an idle bioreactor accessible, but the delay can consume the patient's remaining service window. Reagent transfers and bioreactor relocations also affect which patients can begin production later. Existing work on cell-therapy supply networks and dynamic capacity planning provides the application context [1, 2]. The present study evaluates a synthetic decision model, not a clinically validated prediction system.

The rationale for a graph representation is relational: the value of a request at one site can depend on resources and competing needs at connected sites. A specimen transfer, for example, is useful only if production resources become available before its service window closes. Graph aggregation offers a way to combine a candidate request with related operational states. It does not create new observations or make physical transfers feasible. Its value must be tested against a non-graph model that receives the same permitted information.

A central difficulty is attributing the performance of a composite controller. A learned model can appear effective because it is coupled to a stronger planner, and a graph model can appear effective because its comparator receives less information. Neither comparison isolates the proposed contribution. Cost introduces a second attribution problem: a controller may reduce procurement or other operating expenses while losing more patients. Reporting only the weighted objective can conceal that trade-off.

We therefore ask three separate questions. First, does the complete controller improve on the original operational heuristic? Second, does learning improve on the same planner without a learned correction? Third, does neural message passing improve on a matched representation that retains the same inputs, local feature projection, global readout, planning procedure, and training recipe? We report cost, losses, completions, unfinished service, and adverse worlds for each comparison. An aggregate improvement does not replace these service measures.

The method, identified as Localv2 in the implementation, learns a correction to the cost difference between candidate first actions. It uses paired simulations that run to the original endpoint under a frozen continuation policy. At decision time, a shared eight-step planner supplies candidate scores and the learned correction reranks them. Only the selected first action is executed before replanning. This is simulation-feedback policy improvement with paired Monte Carlo targets. It is not imitation of the planner's selected action, a neural actor, or the earlier temporal-difference state-tail method.

The graph representation uses established graph-convolution operations [3] to combine candidate requests with operational features before producing a scalar cost correction. Its matched Self ablation replaces only neural adjacency with the identity. Both retain the shared planner and global readout. A second, explicitly exploratory study asks whether graph mixing changes the cost–service trade-off when both models instead receive identical bounded neighbor messages and neither uses global H8. This is not a factorial removal of H8 from the first experiment: model, training recipe, and evaluation panel also differ. The two studies therefore answer complementary, not interchangeable, questions about graph value.

The paper makes three contributions within these bounds:

1. A graph-aware candidate-cost controller couples short-horizon planning with paired full-tail simulation feedback under the original twenty-site physical rules. Its graph operations are established components; the specified decision interface and target construction define the method studied here.
2. A five-block comparison separates the integrated operational effect, the learned correction, and the marginal graph effect on the same world panel. The first two yield cost gains; the graph increment over matched Self remains small and unstable.
3. A matched bounded-message study identifies a cross-block graph cost advantage accompanied by increased patient losses. Combined with service accounting and retained negative studies, it distinguishes a cost-only architecture benefit from a patient-protecting system improvement.

The latter finding gives graph representation a substantive role in the analysis without transferring its cost gain to the full Graph/H8 controller or treating patients harmed by the policy as an acceptable unreported expense.

## 2. Related Work

### *2.1. Patient-specific manufacturing and engineering evidence*

Autologous cell-therapy manufacturing connects resource allocation to a patient's treatment pathway. Wang et al. [1] compare centralized and point-of-care supply chains using simulation, assessing cost, fulfillment time, service level, and utilization. Their analysis makes decentralization a conditional design choice, dependent on demand, capacity, and operating rules. Li and White [4] formulate relocatable capacity and supplier-disruption resilience as a Markov decision process. Tseng et al. [5] incorporate changing patient health into manufacturing and scheduling through SimPAC. These precedents motivate joint attention to resources, logistics, and patient condition; they do not establish that a new controller improves clinical outcomes. In the present synthetic problem, patient identity and service windows constrain specimen routing, while reagent transfers and idle-bioreactor relocation remain distinct resource decisions.

Published engineering evidence helps identify cost boundaries and process constraints. Ran et al. [6] separate fixed and variable costs for academic nonprofit CAR-T production, including qualified personnel, equipment, and consumables. Lopes et al. [7] model automation, scale-out, and supply-chain costs; their scenario assumptions are modeled inputs, not observed failure frequencies. Fitzgerald et al. [8] compare manual and automated expansion of adipose-derived stromal cells and analyze labor by production, quality-control, and release roles. That process is not CAR-T, so its labor savings cannot serve as a transferable productivity multiplier. These studies support explicit engineering assumptions and sensitivity questions, not retrospective calibration of experiments completed with different inputs.

Clinical observations require a separate interpretation. Nastoupil et al. [9] retrospectively follow patients undergoing leukapheresis for standard-of-care axicabtagene ciloleucel, including patients who do not reach infusion. Such pathway evidence motivates tracking attrition alongside cost. It neither identifies the simulator's daily patient-loss mechanism nor establishes the causal effect of a scheduling policy. Clinical attrition, modeled manufacturing failure, and synthetic patient loss must therefore remain distinct outcomes.

### *2.2. Learning with operational structure and planning*

Tseng et al. [2] provide a direct reinforcement-learning antecedent for dynamic capacity planning in decentralized PRM supply chains. Inventory re-

search supplies complementary lessons about benchmarking and uncertainty. Gijsbrechts et al. [10] show that carefully tuned deep reinforcement learning can match strong methods for stationary lost-sales, dual-sourcing, and multi-echelon problems, while documenting substantial initial tuning effort. Dehaybe et al. [11] condition inventory policies on rolling demand forecasts and train across randomly generated environments. Their results concern specified inventory models and forecast processes, rather than general immunity to distribution shift. Together, these studies motivate strong operational baselines and explicit statements of which uncertainties a learned policy has encountered.

Hybrid methods retain useful operating structure. Stranieri et al. [12] assign production quantities to reinforcement learning and shipping decisions to multistage stochastic programming in a two-echelon network. Deep Controlled Learning [13] uses approximate policy iteration with sequential halving and common random numbers to allocate simulation effort among candidate actions. Residual reinforcement learning instead superposes a learned action correction on a conventional controller, as demonstrated in robot assembly by Johannink et al. [14]. These are distinct mechanisms. They motivate separating a planner's contribution from a learned increment, but do not imply that a correction to candidate costs is an action residual or that a heuristic safeguard guarantees improvement.

Learning targets also distinguish planning-assisted controllers. Temporal-difference learning updates predictions using differences between successive predictions [15]; DDPG learns a deterministic continuous-action actor and critic [16]. TD-MPC combines short-horizon optimization in a learned latent model with a temporally learned terminal value [17]. The cross-entropy method is an optimization procedure [18], not evidence that a learned component adds value. A correction fitted to simulated full-tail cost differences under a fixed continuation policy differs from these constructions. Its contribution must be evaluated against the same planner without the correction, with candidate generation and evaluation resources held constant.

### *2.3. Graph representations and communication*

Graph convolution provides a structured mechanism for combining node features with adjacency information [3]. In supply chains, Kotecha and del Rio Chanona [19] combine graph state representations with multi-agent reinforcement learning and adapt the parameters of heuristic inventory policies. Graph-aware operational control therefore has established antecedents. Its

relevance here lies in representing interactions among candidate requests, competing queues, and transferable resources, rather than in a claim that combining graph networks with reinforcement learning is itself new. Neither graph representation nor adjacency creates additional observations or changes the physical feasibility of transfers.

Communication and representation must also be distinguished. Tolstaya et al. [20] train decentralized swarm controllers by imitating a centralized expert and exchange information through local, potentially multihop communication. Their study is an imitation-learning precedent for structured communication, not evidence of reinforcement-learning gains in manufacturing. For the present comparison, removing messages tests the contribution of a communication channel. Isolating graph aggregation instead requires a non-graph comparator with the same accessible observations, message content and timing, global summaries, and decision interface. Network-derived planner scores count as information even when no explicit graph message is transmitted.

More expressive aggregation remains a hypothesis for future work. Brody et al. [21] identify the query-independent ranking limitation of standard graph attention and introduce GATv2's dynamic attention. That expressiveness result motivates testing alternative representations; it does not predict an operational gain, patient-service benefit, or general superiority over a matched non-graph model. Changes to aggregation, information access, and planning should consequently be attributed separately.

### *2.4. Decision quality and net service*

Decision-focused learning evaluates predictions through their downstream consequences. Donti et al. [22] train probabilistic models against task objectives in stochastic optimization, including an inventory example. Elmachtoub and Grigas [23] define prediction loss through the decision error induced in an optimization problem. These approaches explain why predictive fit alone need not identify a useful controller. They motivate assessing a candidate-cost model through its selected actions, but do not establish that the present training rule implements either method or inherits their theoretical properties.

The downstream objective also determines what counts as improvement. For patient-specific manufacturing, an operating-cost reduction can accompany fewer completed treatments or more unfinished service. A weighted

objective cannot by itself establish a net service benefit: its component outcomes and accounting horizon must remain visible. Accordingly, cost, patient losses, completions, unfinished service, and adverse scenarios answer different questions. A patient-loss penalty is a modeling choice, not a clinically validated monetary valuation, and an average cost advantage cannot substitute for evidence of patient protection.

### *2.5. Uncertainty, generalization, and statistical limits*

Generalization to broader simulation uncertainty is a substantive open question. Packer et al. [24] distinguish in-distribution evaluation from tests outside the training environment distribution. Forecast-conditioned inventory learning and PRM capacity-planning studies likewise motivate specifying the uncertainty process [11, 2]. For this application, future questions include joint changes in arrivals, disruption dependence, processing times, and observation delays. These are prospective tests, not completed validations. Additional random seeds under one simulator law address sampling variability; they do not establish robustness to a different law or clinical transportability.

Finally, Henderson et al. [25] document variability and reproducibility problems in deep reinforcement learning. Agarwal et al. [26] show how few-run point estimates can support unreliable rankings and advocate interval estimates and distribution-aware summaries. In paired simulation comparisons, uncertainty across independently trained models must remain distinguishable from uncertainty across evaluated worlds; repeated worlds do not create additional independent training runs. Small or unstable graph increments, adverse service trade-offs, and inconclusive comparisons therefore constrain the claims of this study rather than justify a universal ranking of architectures.

## 3. Problem Formulation and Service Accounting

### *3.1. Network and physical constraints*

The benchmark has twenty physical manufacturing sites and $T = 52$ decision epochs. Geographic links, demand generation, production timing, supplier disruptions, resource limits, patient dynamics, and cost coefficients remain those of the original task. The neural representation contains an additional virtual capacity hub, giving 21 representation nodes rather than 21 physical manufacturing sites.

A waiting patient's intact specimen may make at most one qualified direct transfer before manufacturing. Specimens cannot be split, pooled, reassigned, or routed repeatedly. Reagents may move on qualified resource links, and idle bioreactors may relocate on capacity links. An in-process patient batch cannot move. Finished product remains associated with the same patient and returns to the collection site. The primary conditions use a one-epoch specimen-availability lead and zero-epoch finished-product return lead. These are benchmark assumptions, not claims about feasible clinical transport.

Let $x_t$ denote the complete simulator state and $o_t$ the controller's public observation. Public information includes inventories, production and transfer pipelines, observed demand and history, supplier status, patient summaries, and time. The deployed controller cannot use realized future demand, hidden patient attributes, or future random draws. The observation and its encoded features are summaries; we do not claim that they are sufficient statistics for the complete latent state.

### *3.2. Requests and finite-horizon cost*

The controller issues a group-major, eighty-coordinate request,

$$a_t = (a_t^{\mathrm{s}}, a_t^{\mathrm{r}}, a_t^{\mathrm{b}}, a_t^{\mathrm{q}}) \in [-1, 1]^{80}, \qquad a_t^g \in [-1, 1]^{20}. \tag{1}$$

The groups request specimen transfer, reagent transfer, idle-bioreactor relocation, and replenishment. Transfer coordinates express site-level net requests, not arbitrary edge flows. The original decoder enforces availability, quantity, integrality, connectivity, and patient-identity constraints. Consequently, a proposed donor–receiver correction does not guarantee that the native matching rule executes that exact edge movement.

The finite candidate bank restricts the implemented controller's search, not the physical admissibility of the task. All three transfer channels and replenishment remain available. Actions outside the bank were not searched by this method; the experiment does not establish optimal control over the full request space.

The per-epoch cost retains the original decomposition,

$$c_t = c_t^{\text{operating}} + c_t^{\text{patient loss}} + c_t^{\text{expiry}} + c_t^{\text{urgency}}, \qquad r_t = -c_t, \tag{2}$$

and the objective is

$$J(\pi) = \mathbb{E}_\pi \left[ \sum_{t=0}^{T-1} c_t \right]. \tag{3}$$

The discount factor is one. Operating cost already includes charged transfers; those components must not be added a second time. Cost is reported in modeled weighted-objective units, not dollars or realized financial savings. No new settlement phase, changed reward, or extended endpoint is introduced.

### *3.3. Losses, completions, and unfinished service*

Let $L$ be cumulative patient losses, $D$ completed services, and $U_T$ active unresolved patient obligations at the endpoint. Losses include the simulator's recorded expiry and ineligibility events; completions require the modeled patient service to finish. Unresolved service includes waiting patients, patients in production, specimens in transit, and any active finished-product or return obligations. It is not just the waiting queue. The recorded `cutoff.terminal_patients` field includes resolved outcomes and must not be substituted for $U_T$.

For a panel with identical enrollment $N$ and exhaustive endpoint states, patient accounting gives $N = L + D + U_T$. Thus candidate-minus-reference differences satisfy

$$\Delta D = -\Delta L - \Delta U_T. \tag{4}$$

This accounting identity does not make any one outcome a sufficient service criterion. Fewer losses can coexist with more unresolved obligations, and a smaller queue can result from losses rather than successful service. We retain all three outcomes and their observation horizon.

## 4. Planning-Assisted Candidate-Cost Learning

### *4.1. Shared H8 planner and structured candidates*

The shared planner, denoted H8, considers at most fourteen first-action candidates. A deterministic generator uses public state and static physical topology to construct original MDL2 and MDL3 anchors and structured corrections. The corrections cover specimen movement, reagent movement, idle-capacity movement, replenishment, and complementary resource changes. Shortlisting uses public shortage, urgency, logistics, and diversity rules. It is not a trained proposer or a search over arbitrary joint action sequences.

Candidate deduplication can reduce the bank size. It preserves equivalent anchor behavior and does not refill an empty family's quota with more actions from another family. The physical decoder, rather than the shortlist label,

determines realized flow. This generator differs from the earlier fixed epsilon-template bank; the later results must not be attached to that older candidate specification.

For $a \in \mathcal{A}(o_t)$, H8 evaluates the first request followed by original MDL2 continuation in two sampled prediction futures for $h_t = \min(8, T-t)$ epochs:

$$H_t(a) = \frac{1}{2} \sum_{k=1}^{2} \sum_{j=0}^{h_t-1} \widehat{c}_{t+j}^{(k)}(a, \mathrm{MDL2}). \tag{5}$$

There is no learned terminal value in this score. Prediction futures use the existing public-information demand estimator and conditional patient sampler, not the realized evaluation future. The same prediction context is reused across candidates at a decision. Shared stream identities do not, by themselves, prove that every modeled event remains pathwise aligned after decisions diverge.

Let $a_0$ be H8's selected anchor, including its existing tie rule. A learned model supplies a correction to each candidate relative to this anchor:

$$S_\theta(o_t, a) = H_t(a) + s\{f_\theta(X_t(a)) - f_\theta(X_t(a_0))\}, \qquad s = 10^9. \tag{6}$$

The controller selects the smallest corrected cost and retains the anchor when the improvement is no greater than $10^{-6}$ native-cost units. That tolerance is a numerical selection rule, not a patient-safety margin. The system executes only the selected first action, observes the next state, and replans.

### *4.2. Paired Monte Carlo targets*

During a collection round, the continuation policy $\pi_r$ is frozen. At each selected training state, the collector compares the H8 anchor with a distinct alternative selected by the fixed collection rule. Each first action is followed by $\pi_r$ through the original endpoint. Two native futures per action supply paired full-tail costs. Denote their mean by $\overline{G}_{\pi_r,t}(a)$. The fitted target is

$$y_r(a, a_0) = \frac{\{\overline{G}_{\pi_r,t}(a) - \overline{G}_{\pi_r,t}(a_0)\} - \{H_t(a) - H_t(a_0)\}}{s}. \tag{7}$$

Lower values favor the alternative. The pair compares complete costs under one declared continuation policy; it is neither a reward-sign advantage nor a full action-value target added again to H8.

The learner minimizes centered mean squared error on the current round's pairs:

$$\mathcal{L}_r(\theta) = \frac{1}{|\mathcal{B}|} \sum_{(a,a_0,y)\in\mathcal{B}} \left[f_\theta(X_t(a)) - f_\theta(X_t(a_0)) - y\right]^2. \tag{8}$$

The anchor output is centered by subtraction, not fitted as a separate absolute cost. Prior-round buffers are not replayed. After the fixed updates, the new model defines the next collection policy.

This procedure learns from simulated consequences of its actions rather than from labels telling it to imitate H8. It is a planning-assisted form of simulation-feedback policy improvement. Unlike the earlier TD8 state-value method, it uses no bootstrapped learned terminal target in Eq. (7); all tails reach $T$. It also does not train a DDPG, PPO, or SAC actor. References to a learning increment in this paper concern the specific Graph/H8 versus H8 comparison, not all reinforcement-learning methods or online deployment adaptation.

Training simulator snapshots and future outcomes supply labels and pairing provenance only. They are not predictor features. Evaluation disables fitting and does not expose those training-only outcomes to the controller.

### *4.3. Graph mechanism and matched Self representation*

Both learned models receive $X_t(a) \in \mathbb{R}^{21\times 41}$. The first 36 columns are the same public feature encoding. Four columns add the site's specimen, reagent, idle-bioreactor, and replenishment requests; the virtual hub receives zeros in these four columns. The final column repeats $\{H_t(a) - H_t(a_0)\}/s$ across nodes. This score is a public-information planning output, not an observed future cost.

The encoder graph merges the configured information, specimen, reagent, and capacity relations into one undirected adjacency. In this representation, capacity links use a virtual hub connected to all twenty sites; the hub carries aggregate idle and total capacity, time, and its indicator. It is not a physical transfer station. Physical transfer eligibility remains governed by the simulator, independently of this computational graph. Facility-to-facility weights decrease with configured geographic travel time, with a positive floor; hub links have unit weight. Duplicate relations are merged rather than assigned different learned message functions.

Writing the weighted adjacency as $A$, the encoder uses $\widehat{A} = \widetilde{D}^{-1/2}(A + I)\widetilde{D}^{-1/2}$, where $\widetilde{D}_{ii} = \sum_j (A + I)_{ij}$. The graph backbone uses two width-64

layers. With the prescribed symmetric normalized adjacency $\widehat{A}$ including self-loops,

$$Z^{(0)} = X, \qquad Z^{(\ell+1)} = \mathrm{ReLU}\Big(\widehat{A}[Z^{(\ell)}W_\ell + \mathbf{1}b_\ell^\top]\Big), \quad \ell = 0, 1. \tag{9}$$

Localv2 adds a learned raw-feature projection before a flattened global readout:

$$Z = \mathrm{ReLU}\big(Z^{(2)} + XW_{\mathrm{raw}} + \mathbf{1}b_{\mathrm{raw}}^\top\big), \qquad f_\theta(X) = w^\top \mathrm{vec}(Z) + b. \tag{10}$$

The scalar head is initially zero, so the initial corrected policy follows H8.

For a given candidate, each graph layer mixes transformed state and request features before applying its nonlinearity. This permits a site's representation to depend on its encoded neighbors' circumstances. The raw-feature skip retains a direct local path, while the global head combines all node outputs. Two layers also permit site-to-site paths through the hub. The model is therefore neither a strictly local controller nor a learned communication protocol. Its proposed advantage is a relational processing bias, not privileged or faster information. The aggregate adjacency also does not separately model reagent, specimen, and reactor interactions; successful cost prediction is not proof that any one physical mechanism was learned.

Graph and Self each have 10,881 trainable parameters. Self replaces only $\widehat{A}$ with $I_{21}$; its local projection, flattened head, feature inputs, physical links, and shared planner remain unchanged. Within a training block, trainable tensors and the initial optimizer and private batch-generator state match. Physical isolation is not part of this ablation. Self can still combine site information through its global readout and exploit the shared planner. The comparison therefore estimates the marginal value of neural message passing within this full controller.

The global head in Eq. 10 is linear. With identity adjacency, Self is a sum of nonlinear node-wise terms at the encoded-feature interface, not an arbitrary nonlinear MLP over concatenated raw sites. Conversely, global planning features can already encode joint decisions before this head. The encoder has 36 facility links and 20 hub links, not a complete graph of 190 direct facility links. Two-hop reachability does not make Graph and Self functionally equivalent. The fixed 21-node flattened readout also does not provide arbitrary-size invariance.

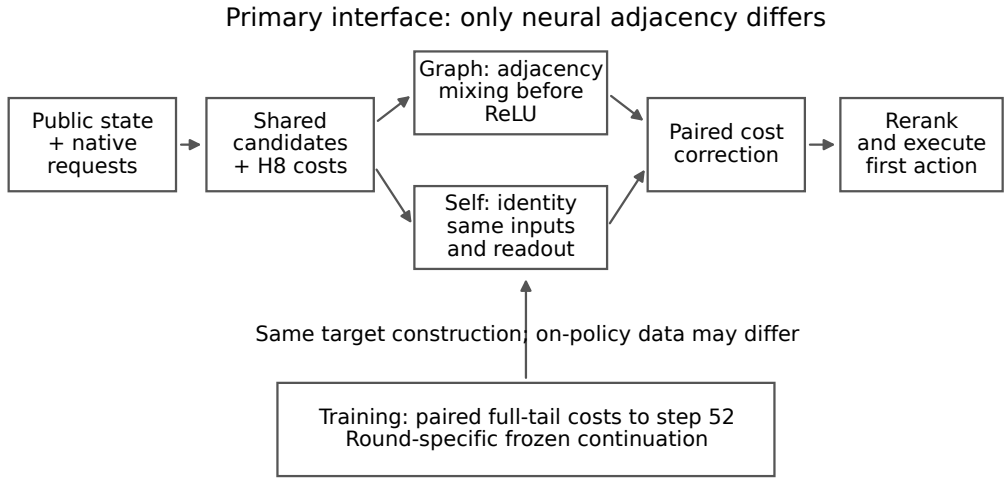



Figure 1: Primary controller and attribution interface. Graph and Self use the same feature, target, and planning definitions; neural adjacency differs. Their on-policy training trajectories may diverge, so the realized training examples are not identical. Evaluation freezes all models and executes only the selected first action. The separate bounded-message experiment uses a different interface and cannot be represented by deleting H8 from this diagram.

### *4.4. Training rounds and frozen evaluation*

The replication trains five Graph/Self pairs independently of the earlier three-block panels. Each model collects 96 main training worlds in three rounds of 32, using the fixed round-robin condition order. The collection recipe selects three states per world and retains its original alternative-selection, deduplication, and no-distinct-alternative skip rules. After each round it makes 1,024 Adam updates with 64 pairs per batch and learning rate $10^{-4}$. Across ten models, the completed training record contains 960 main trajectories and 30,720 updates. Counterfactual tails and planner predictions are additional computation, not additional independent training worlds.

Matching initial tensors and stream allocations does not guarantee identical on-policy data. Once decisions diverge, the two models can visit different states and collect different pair-buffer sizes; subsequent batch indices can also differ. The experiment preserves matched initialization and the same sampling rule, not a false claim of identical realized samples throughout training. All ten final models are sealed before evaluation. No intermediate checkpoint

Table 1: Three routes by which network information enters the primary controller. Only the last row changes in the Graph/Self ablation. No route exposes the realized evaluation future.

| Route | Shared content or architectural difference |
|---|---|
| Candidate construction | Both methods use public state and physical topology to construct the same type of structured request bank. |
| H8 and public features | Both receive the planner's scores, public node features, capacity-hub summaries, and candidate request coordinates. |
| Neural processing | Graph mixes neighboring features with $\widehat{A}$ before the nonlinearities; Self uses $I$. Both retain the raw-feature skip and global scalar readout. |

is selected on evaluation performance, and evaluation performs zero optimizer updates.

## 5. Experimental Design

### *5.1. One panel and three attribution questions*

Table 2 defines the four evaluated methods. Graph/H8 versus MDL2 measures the integrated operational effect; Graph/H8 versus H8 measures the effect of adding the learned correction; Graph/H8 versus Self/H8 measures the marginal neural-adjacency effect. Self/H8 versus H8, Self/H8 versus MDL2, and H8 versus MDL2 complete all six pairwise comparisons. Historical Flat comparisons remain separate; a new Flat model was not added to this replication or substituted for Self after results were observed.

Five training blocks, four fixed conditions, and six paired worlds per block–condition cell give 120 worlds and 480 complete evaluations. Conditions are nominal history (0), abrupt regime shift (1), regional drift (2), and compound regional stress (3). A world is paired across all four methods. Training randomness has five blocks, not 120 independent replications. Older selected panels are not pooled into these estimates.

The completed evaluation used a separately authorized recovery after the original attempt failed. It reused the ten sealed final models and all 960 completed training trajectories, without new training, labels, seeds, or seals. The original failed attempt remains in the record. The recovery's completed

**Algorithm 1** Planning-assisted paired cost-correction learning

```
Initialize Graph or Self with a zero-output correction head
for each of the three fixed policy rounds do
    Freeze the current continuation policy
    Collect 32 worlds in the fixed condition order
    At prescribed states, form an H8 anchor and distinct alternative
    Run paired native futures through T under frozen continuation
    Form current-round residual labels using Eq. (7)
    Perform 1,024 centered-MSE updates using Eq. (8)
end for
Seal the final model before any evaluation
for each evaluation decision do
    Form public structured candidates and shared H8 scores
    Rerank with Eq. (6); retain the anchor on a tie
    Execute the first request only, with learning disabled
end for
```

comparison and locally verified archive define the main result panel; later procurement preservation is a different experiment.

### *5.2. Metrics, uncertainty, and fixed screens*

For candidate $c$ and reference $r$, reported cost saving is

$$R_{c/r} = 100\,\frac{\overline{C}_r - \overline{C}_c}{\overline{C}_r}. \tag{11}$$

It is the difference of mean costs divided by the reference mean, not the mean of world-level percentages. Different contrasts have different denominators, so their percentages are not additive. Patient deltas are candidate minus reference: positive $\Delta L$ is adverse, positive $\Delta D$ is favorable, and positive $\Delta U_T$ is adverse.

The saved analysis uses 2,000 paired hierarchical bootstrap resamples. It samples five training blocks with replacement, keeps the four conditions equally weighted, and samples six paired worlds within each selected block–condition. The same resampled indices apply to all methods and contrasts. The percentile 95% intervals are descriptive and not adjusted for multiplicity. They are not confirmatory significance tests or clinical noninferiority intervals.

Table 2: Methods in the same five-block evaluation. G, S, H, and M are table abbreviations used throughout. All retain the original physical task.

| Code | Method | Decision rule and interpretation |
|---|---|---|
| M | MDL2 | Original operational heuristic; no learned correction. |
| H | H8 | Structured candidates, two prediction futures, eight-step planning and MDL2 continuation; no learned tail or correction. |
| S | Self/H8 | H8 plus paired candidate-cost correction; identity neural adjacency, local projection and global readout. |
| G | Graph/H8 | Same inputs, parameter count and training recipe as S; graph neural adjacency. |

The fixed cost screen requires at least 0.5% mean saving, positive saving in every block, and a positive descriptive lower cost bound. For the primary Graph/Self and Graph/H8 comparisons, patient protection additionally requires nonpositive mean extra losses overall and in every condition, plus a nonpositive descriptive upper patient-loss bound. The other saved comparisons retain their recorded overall/condition mean patient guardrails. We reproduce these screens as implemented rather than impose a new rule retrospectively. Completions and unresolved service remain reported outcomes even where they are not components of a particular fixed screen.

### *5.3. Computational and preservation accounting*

Training, native evaluation, planning predictions, neural scoring, and elapsed time are separate quantities. The evaluation recovery records 24,960 native steps, 1,450,576 prediction steps, and 12,480 learned-model forward calls, with zero evaluation updates. These are execution counts, not measures of statistical sample size. The prediction burden can differ despite identical horizon and candidate ceilings because states and deduplicated banks differ.

The complete evaluation phase took 3,217.136 seconds in the recorded run, and analysis/archive took 534.398 seconds. Table 6 gives saved method-level totals; summed trajectory times are not the wall time of the parallel evaluation. No hardware-independent speedup or computational graph advantage is inferred from these timings. The main local archive is recorded as verified; this does not establish off-device backup, cloud synchronization, or public availability.

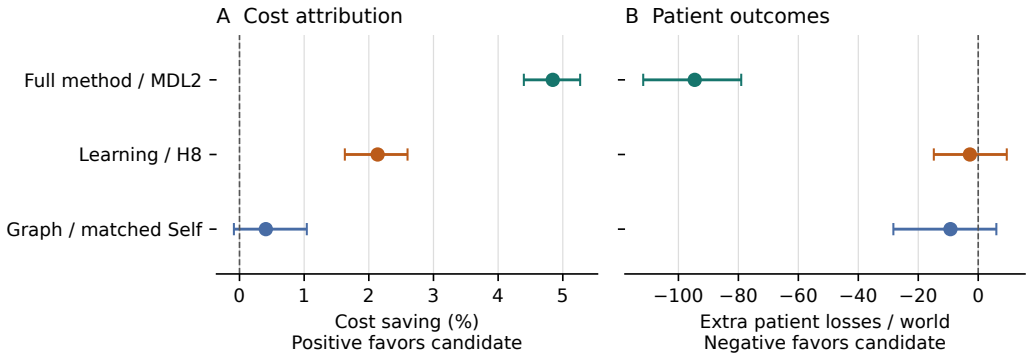



Figure 2: Three attribution questions on the same five-block, 120-world panel. Points and bars are saved means and descriptive, unadjusted 95% intervals. Positive cost saving favors Graph/H8; negative extra losses favor Graph/H8. The learning-related cost result does not imply patient protection, and the Graph/Self interval includes zero. These intervals are not multiplicity-adjusted confirmation after the development search.

## 6. Results

Figure 2 separates the three primary contrasts. All figures in this revision display saved comparison fields; no intervals or scientific comparisons were recomputed for the manuscript.

### *6.1. Overall operational benefit*

Graph/H8 reduces mean cost relative to MDL2 by 4.84499% (Table 3), with 94.567 fewer losses and 110.942 more completions per world (Table 4). All five block cost means favor Graph/H8, and the saved operational joint screen passes. Its average unresolved service is also lower (Table 5). These results support the integrated controller in this development panel.

The improvement is not specific to graph learning. H8 alone reduces cost by 2.768% relative to MDL2, with 91.792 fewer losses and 91.767 more completions. Self/H8 also improves on MDL2. These comparisons show why the complete Graph/H8–MDL2 difference cannot be attributed wholly to the graph operator or to learning. Moreover, 16 of 120 worlds have more losses under Graph/H8 than MDL2 despite its favorable average and operational screen.

### *6.2. Incremental learning benefit*

Relative to the shared H8 planner, Graph/H8 reduces mean cost by 2.13645%, with a descriptive interval of $[1.629, 2.599]$%. Cost is lower in

Table 3: All six saved cost contrasts on 120 paired worlds. Positive saving favors the candidate. Intervals are descriptive 95% intervals; harmed worlds have higher candidate cost. The last column is the recorded joint development screen, not a clinical guarantee.

| Candidate/reference | Saving (%) | Interval (%) | Harmed | Joint |
|---|---|---|---|---|
| G/M | 4.845 | [4.398, 5.268] | 6 | Pass |
| G/H | 2.136 | [1.629, 2.599] | 27 | Fail |
| G/S | 0.407 | [-0.087, 1.041] | 51 | Fail |
| S/H | 1.736 | [0.928, 2.417] | 36 | Fail |
| S/M | 4.456 | [3.642, 5.105] | 9 | Pass |
| H/M | 2.768 | [2.297, 3.242] | 10 | Pass |

Table 4: Patient outcomes for the same six contrasts. Deltas are per-world means. The interval applies to extra losses. $n_L$ counts worlds with more losses, and $n_D$ counts worlds with fewer completions, each out of 120. These harm categories need not identify the same worlds.

| Contrast | $\Delta L$ | Loss interval | $\Delta D$ | $n_L$ | $n_D$ |
|---|---|---|---|---|---|
| G/M | -94.567 | [-111.744, -79.050] | +110.942 | 16 | 15 |
| G/H | -2.775 | [-14.809, 9.528] | +19.175 | 63 | 51 |
| G/S | -9.258 | [-28.298, 6.052] | +9.550 | 53 | 49 |
| S/H | +6.483 | [-12.451, 30.926] | +9.625 | 68 | 60 |
| S/M | -85.308 | [-108.492, -59.474] | +101.392 | 19 | 18 |
| H/M | -91.792 | [-107.577, -77.466] | +91.767 | 4 | 4 |

all five block means. This is evidence for a learning-related cost increment under the specified planning interface.

The service conclusion is narrower. Mean extra losses are $-2.775$, but their interval is $[-14.809, 9.528]$, and 63 of 120 worlds have more losses. Abrupt regime shift and regional drift have mean extra losses of +4.200 and +26.033, respectively. Regional drift also has 11.033 fewer completions. Consequently, the patient and joint screens fail. More completions on average do not establish protection in every condition or world.

Self/H8 likewise reduces cost relative to H8, by 1.736%, while increasing mean losses by 6.483. Its 9.625 additional average completions coexist with increased losses in 68 worlds and fewer completions in 60. Both learned models can improve the weighted objective without satisfying the desired patient protection.

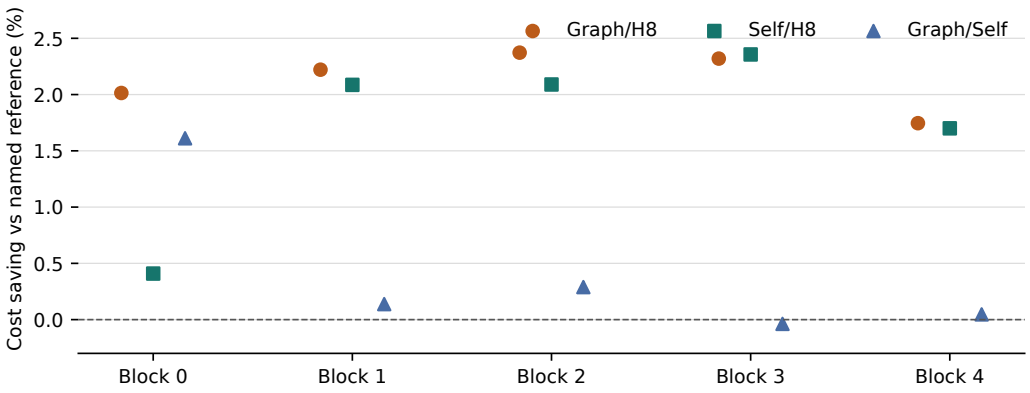



Figure 3: Parallel block-level comparisons requested during collaborator review. Each block contains 24 paired worlds. Both learned controllers improve the shared H8 cost in all five block means, while Graph/Self is dominated by block 0. Each series uses its named reference; the three percentages are not additive.

### *6.3. Marginal graph contribution*

Graph/H8 costs 0.40707% less than Self/H8 on average. The descriptive interval $[-0.087, 1.041]\%$ crosses zero, the mean misses the fixed 0.5% threshold, and one block has higher graph cost. The five block savings are 1.611%, 0.138%, 0.289%, −0.038%, and 0.046%. The effect is therefore small and uneven relative to the prespecified replication criterion.

Graph/H8 has 9.258 fewer losses and 9.550 more completions on average than Self/H8, but its extra-loss interval reaches +6.052 and 53 worlds have more losses. All four condition mean loss differences favor Graph/H8; this does not remove the block and world heterogeneity. The cost, patient, and joint screens fail. These observations neither establish independent graph superiority nor show that all graph representations are ineffective.

Figure 3 makes the training variability explicit. Four blocks are near ties; most of the pooled Graph/Self difference comes from block 0, where Self's H8-relative improvement is smaller. This is compatible with a small graph effect or a difference in trainability. Five blocks do not identify one explanation, and they do not establish that a single faulty Self seed caused the result. Every condition-specific Graph/Self cost interval also includes zero.

Table 5: Saved mean patient accounting at epoch 52, per world. Pending is all active unresolved service, not only waiting. Other active finished/return obligations are zero in these saved means. Rounded cells need not sum exactly.

| Method | Lost | Completed | Pending | Waiting | Production | Transit |
|---|---|---|---|---|---|---|
| G | 2331.333 | 2976.858 | 674.792 | 540.867 | 127.067 | 6.858 |
| S | 2340.592 | 2967.308 | 675.083 | 542.600 | 125.667 | 6.817 |
| H | 2334.108 | 2957.683 | 691.192 | 563.450 | 123.633 | 4.108 |
| M | 2425.900 | 2865.917 | 691.167 | 567.525 | 120.267 | 3.375 |

Table 6: Saved evaluation computation for 120 trajectories per method. Forward calls batch candidate examples. Trajectory elapsed totals are summed durations, not parallel wall time or a controlled latency benchmark.

| Method | Prediction steps | Forward calls | Scored examples | Elapsed (s) |
|---|---|---|---|---|
| G | 512816 | 6240 | 34557 | 5756.741 |
| S | 503918 | 6240 | 34038 | 5675.141 |
| H | 433842 | 0 | 0 | 4996.719 |
| M | 0 | 0 | 0 | 65.563 |

### *6.4. Patient harms and endpoint obligations*

The aggregate service ledger in Table 5 distinguishes successful completion, loss, and unresolved service. Waiting constitutes most unfinished service, but production and transit remain material obligations. For example, H8's substantial reduction in losses relative to MDL2 does not coincide with a lower mean endpoint pending count. This illustrates why loss reduction alone does not settle the service question.

Appendix A retains all six contrasts by block and condition, including the cost-, loss-, and completion-harm counts. Individual paired-world records remain in the source comparison; no world has been dropped or replaced to improve a manuscript estimate. Even a condition with favorable mean losses may contain harmful worlds. The operational mean improvements, learning-related cost improvement, and uncertain graph effect are therefore compatible findings, not competing descriptions of a single universal gain.

## 7. Graph Value Under Matched Bounded Communication

### *7.1. A separate matched-information test*

A later experiment replaces the shared global H8 interface with a synthetic one-round message interface. Its graph and non-graph message models each have 30,017 parameters and receive matched messages, static relations, initial tensors, training data, and batch indices. It uses three training blocks and 24 paired worlds, not the five-block primary panel. The original Localv2 models and H8 are unchanged by this experiment.

Each site keeps its own observation and receives twelve operational summaries from each permitted neighbor, covering demand, waiting patients, inventories, capacity, supplier status, forecasts, pipelines, and waiting-patient urgency and transfer eligibility. The same fixed permission graph has 36 undirected links. One communication round sends 72 packets of 64 bytes, or 4,608 modeled bytes per cycle. The protocol additionally charges a 65,536-byte static provisioning envelope per role and world; these are modeled communication allowances, not measured network traffic. Both models receive the same static relation rows and the same transcript on an identical state. There is no relaying, dynamic hub, global H8, MDL2 advice, or access to unreceived remote dynamic fields. Physical links are not removed by the communication restriction.

The two-layer message GCN mixes the permitted ego-graph features. Its matched non-graph model uses identity mixing but retains a nonlinear processor of its own features and the received-neighbor mean. Both have the same raw-feature skip and nonlinear readout. Two computational layers do not imply two message rounds. Six models are fitted to the same existing 96 collection worlds, with 1,024 updates per model and matched initialization and batch indices. The messages affect scoring, not the own-site candidate bank. Selected site requests are assembled and executed by the unchanged physical matching rule. Thus the comparison changes graph processing, not information access, candidate availability, or communication budget; equal parameter counts do not establish equal execution time.

### *7.2. Cross-block cost advantage with patient harm*

The message GCN reduces cost by 4.843% relative to the message MLP, with a descriptive interval of $[3.417, 6.101]$%. All three block means and all four condition means favor the GCN in cost (Table 7). Only one of 24 worlds has higher GCN cost. This is stronger cross-block cost evidence within its

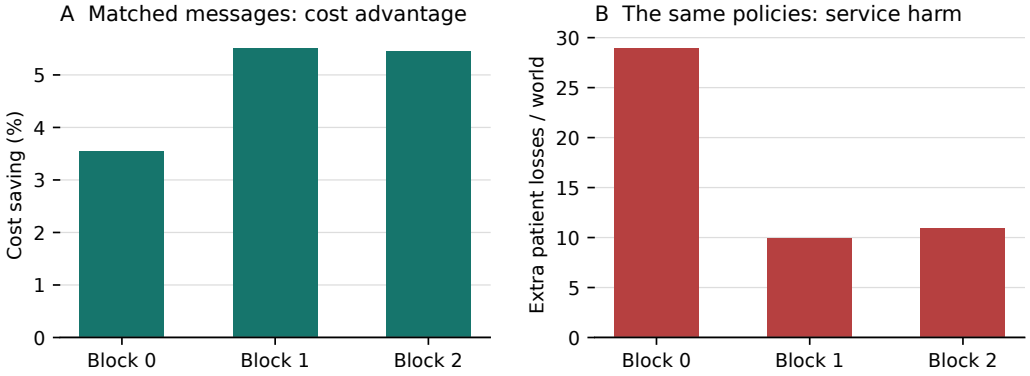



Figure 4: Cost and service must be read together in the bounded-message study. The same-information GCN has lower mean cost in each of three blocks but higher mean patient loss in each block. Each block contains eight paired worlds. Neither pooled uncertainty nor a cost-only screen establishes service equivalence.

Table 7: Message GCN relative to the matched message MLP in the separate bounded-communication study. Positive saving favors GCN; positive loss delta is adverse. Block rows contain eight paired worlds each. Aggregate values are transcribed from the saved comparison, not averaged from rounded rows.

| Training block | Cost saving (%) | Extra losses per world |
|---|---|---|
| 0 | 3.543 | +29.000 |
| 1 | 5.520 | +10.000 |
| 2 | 5.454 | +11.000 |
| All 24 worlds | 4.843 | +16.667 |

own protocol than the small Graph/Self increment in the primary panel; it is not a statistical comparison of the two experiments' effects.

The patient result reverses that favorable interpretation. The GCN loses 16.667 more patients and completes 12.375 fewer services per world. Fourteen of 24 worlds have more losses, and losses increase in every block mean. The four condition loss differences are $-9.000$, $+69.333$, $+21.667$, and $-15.333$. The descriptive extra-loss interval is $[-19.126, 55.769]$; its inclusion of zero is not evidence of patient equivalence. The fixed joint screen fails. The supported finding is a cost-only graph-architecture signal under matched finite messages, not a patient-protecting improvement.

The retained Local32 comparison changes both message access and architecture. Message GCN costs 5.530% less and loses 63.042 fewer patients on average than Local32, but block 0 costs 4.128% more and block 2 loses 89 more patients. Message MLP costs 0.722% less and loses 79.708 fewer patients on average than Local32. These secondary system comparisons do not isolate graph mixing and do not replace the matched message GCN/MLP result.

### *7.3. What the two interfaces do and do not establish*

The main panel tests a graph bias on top of a shared network-aware planner and global readout. The message study tests graph mixing on identical bounded transcripts without that global planner. Neither gives GCN more observations than its matched comparator. The latter cost signal warrants reporting graph structure as an empirically consequential design choice in that protocol, while its patient harms prevent calling it an overall superior policy.

The numeric proximity between this 4.843% result and the primary 4.845% Graph/H8–MDL2 result is incidental. They use different policies, baselines, information interfaces, training blocks, and world panels. Neither percentage can be substituted for the other or combined with the other's service outcomes. The imposed message limit is also a synthetic design choice, not a measured constraint from a manufacturing organization.

The saved cost components locate an important trade-off (Fig. 5). Lower reagent procurement accounts for 200.348 million simulated cost units of saving per world, while patient-loss cost increases by 8.333 million and bioreactor-shortage cost increases by 5.088 million. These are accounting differences between realized policies, not causal effects of individual actions or actual monetary savings. They motivate a procurement audit, but do not prove that removing excess purchases will remove the cross-architecture service gap.

These studies do not isolate information restriction or H8 as the cause of the difference in graph effects: they also change model, training, action interface, and world panel. They therefore cannot establish that global H8 suppresses graph learning or that the original Self was unfairly strong. The bounded-message targets also describe a single-site action change under the original frozen Local continuation, whereas deployment assembles multiple sites' new decisions. That mismatch is a possible contributor to service errors, not an identified explanation of the graph/non-graph gap.

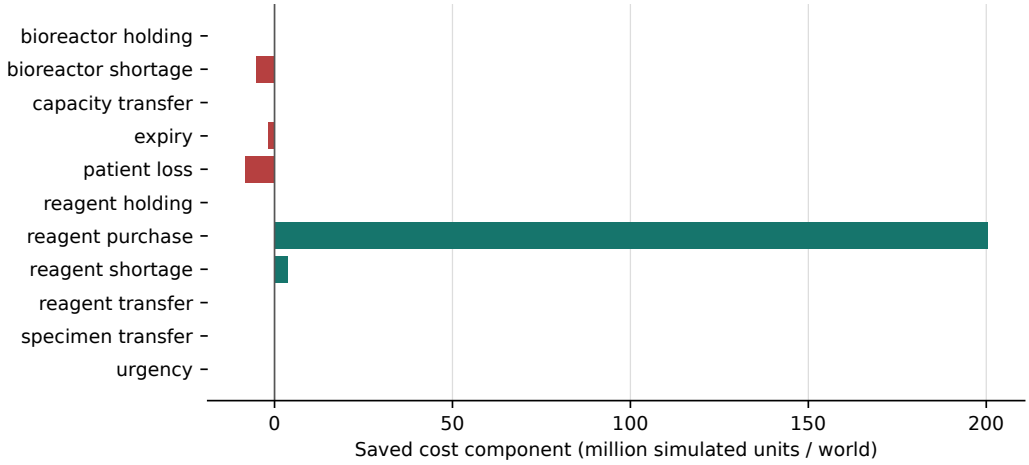



Figure 5: Saved component-level cost savings of message GCN versus message MLP. Positive values are lower GCN cost; negative values are higher GCN cost. Reagent procurement dominates the total saving, while patient-loss and some resource penalties worsen. The common scale retains small components rather than suppressing unfavorable ones. Units are synthetic accounting units.

## 8. Service Preservation and Negative Development Evidence

### *8.1. Procurement preservation within each policy*

A separate procurement study applies the same conservative cap to each frozen message architecture. Its completed reexecution contains 24 paired worlds and 96 evaluations, comparing capped and uncapped policies within each architecture and across architectures. The saved reconciliation reports preservation of each architecture's own patient and physical trajectory, including the paired-state checks, while procurement accounting changes.

Within the GCN policy, the cap reduces mean cost by 9.186% with zero additional losses and zero additional completions. Within the non-graph policy, it reduces cost by 10.645%, also with zero changes in those outcomes. The larger gain belongs to the non-graph policy. These are common-rule procurement gains, not new reinforcement-learning gains or graph-specific improvements.

With both architectures capped, the GCN still costs 2.940% less but loses 37.625 more patients and completes 36.333 fewer services per world. Eighteen of 24 worlds have more losses. Its block loss differences are −28.500, +84.500,

and +56.875. Thus preserving each policy relative to itself does not make the two policies equivalent in patient service.

The separate archive-only recovery closed local preservation without changing these estimates. Failure and closeout records remain distinct in Appendix D; no training or evaluation was repeated for this closeout.

### *8.2. Negative development screens*

Several bounded development studies tested explanations or remedies for the remaining graph and service gaps. They are not pooled with the primary panel. Table 8 summarizes their recorded decisions. A failed implementation attempt is distinguished from a completed negative screen; only completed comparisons support performance statements.

A procurement/allocation crossover worsened its own GCN baseline when GCN procurement was combined with non-graph allocation. Deadline coordination improved each family's own losses but retained the matched cross-family service gap and increased pending service. The feasible-start selector had limited action exposure and also retained the cross-family gap. Competitive allocation reduced mean losses while increasing unfinished service and harming one condition. The joint-service rule changed no action, so its baseline differences cannot be credited to the intervention.

A training-only feasibility study subsequently examined direct loss and pending-service targets without fitting a new model. Loss effects showed some repeated-future stability, whereas the pending-service signal was weak and inconsistent across blocks. This did not support preparation of a target-only fit. It did not estimate conditional predictability from all available inputs, compare new architectures, or prove that service learning is impossible.

The later common-joint-action comparison also failed to establish an architecture gain: under matched information and a common continuation, GCN cost was 0.0184% higher, losses were 0.980 higher, and completions were 0.830 lower than the non-graph model. The GCN cost and service differences were unfavorable in every block and condition mean. A representation-factorial screen then localized improvements to one block rather than establishing a cross-block repair. Neither supports adding a more complex encoder by default.

A conditional-recipient screen fitted nine short models and reused twelve already-viewed development states, rather than evaluating complete new policies. Relative to its non-graph comparator, the GCN increased cost

Table 8: Retained negative or inconclusive development directions. These are distinct protocols, not additional replications of the main panel. Their sources and interfaces remain separate in the local evidence record.

| Direction | Recorded outcome and scope of inference |
|---|---|
| More future samples | The matched mean-two/mean-four precision screen failed in all blocks; more labels alone did not establish a decision repair. |
| Complete-candidate decision loss | Graph lost to Self in every block; both families worsened relative to their own mean-squared-error cells. |
| Manufacturing auxiliary targets | Small pooled effects did not pass cross-half direction and patient safeguards. |
| Conditional H8 queries | Direction changed across orientations; quality and mechanism gates failed. |
| Procurement and allocation crossover | The prespecified hybrid worsened its own GCN cost and patient outcomes; the reverse hybrid is not a replacement winner. |
| Deadline and feasible-start rules | Own-policy improvements did not establish matched cross-architecture service protection; pending burden or low exposure remained. |
| Competitive allocation | Lower average losses coexisted with increased unresolved service and condition-level harm. |
| Joint-service rule | No action changed; an inactive rule cannot explain baseline performance. |
| Direct loss/pending targets | Existing training labels did not support a stable pending-service constraint across blocks; no new model was fitted. |

by 0.0296% while reducing losses by 2.063 and increasing completions by 2.333; GATv2 increased cost by 0.0477%, reduced losses by 1.292, and increased completions by 1.458. The fixed screens failed. Lower candidate regret in this small menu did not translate into the required cost–service result. Subsequent training-data inspection found no stable missed beneficial forced choice in the viewed development menu. A multi-resource inventory found public reagent/bioreactor co-pressure in zero, eight, and zero of eight inspected roots across the three blocks. This narrow support does not justify claiming a cross-block mechanism or automatically starting another fit.

## 9. Discussion and Limitations

### *9.1. What the comparisons establish*

The integrated method improves on MDL2 in mean cost and patient outcomes on the primary development panel. H8 and Self/H8 also improve on MDL2, so the full effect is a property of the controller and its comparison, not a direct estimate of the value of graph learning. Graph/H8 improves the shared H8 cost in every block, but its condition-level patient harms prevent an unconditional service claim. The graph-versus-Self point estimate is favorable and small, with uncertainty and block disagreement that fail the fixed screen.

### *9.2. Graph structure as a conditional design choice*

The matched message experiment provides a positive graph finding that should not be erased by its failed joint screen: GCN lowers cost in all three block means under the same permitted messages, parameter count, and fitting opportunity. It also increases mean losses in all three blocks. Graph structure can therefore change the achieved cost–service trade-off in this protocol without producing a patient-protecting improvement. Neither "graph structure is useless" nor "graph structure makes the system better overall" follows from these results.

In the primary controller, shared H8 and the flattened global readout may leave little additional value for neural message passing. This is a plausible explanation, not an identified mechanism or a reason to deprive Self of information. The comparison holds the intended architecture contract fixed while allowing on-policy data to differ. It does not isolate the graph operator under identical realized training examples, and the observed gap can include the downstream consequences of different policies.

The practical question is consequently narrower than whether a GCN can beat an information-poor baseline: can a relational representation improve decisions over a matched information processor while retaining service performance? The present studies establish a cost-only signal in one interface and a small, uncertain increment in the other. A new representation or hybrid decision mechanism would need its own matched comparison; it cannot inherit the primary controller's patient gains or the message model's cost reduction.

Table 9 states the supported graph claims at the interfaces where they were measured. These are observed protocol-specific results, not identified causal conditions under which GCN must succeed. In particular, bounded

Table 9: Positioning the graph contribution without transferring evidence between protocols. Loss deltas are candidate minus reference per world.

| Interface and comparison | Supported finding and boundary |
|---|---|
| Shared H8: Graph/Self | 0.407% lower cost, interval crosses zero, block disagreement; independent graph superiority remains unestablished. |
| Matched bounded messages: GCN/MLP | 4.843% lower cost with all three block cost means favorable, but +16.667 losses and −12.375 completions; a cost-only architecture signal. |
| Both procurement-capped: GCN/non-graph | 2.940% lower cost but +37.625 losses and −36.333 completions on a separate panel; own-policy preservation does not establish service equality across architectures. |
| Symmetrically stale advice: Graph/Self | 0.150% lower cost, interval crosses zero and patient safeguard fails; the 9.026% advantage over stale H8 does not isolate message passing or faster communication. |

information, a different fitting recipe, and a different action interface changed together between studies.

### *9.3. Why cost and patient service diverge*

The objective combines several components and is not a hard patient-protection constraint. A policy can reduce one operating component while changing which patients receive timely resources. Conversely, a public rule can remove unneeded procurement without changing that policy's patient path. The latter is useful engineering evidence, but it cannot repair a service difference that already exists between two policies.

Endpoint accounting is equally important. A patient who remains active at epoch 52 has neither completed service nor necessarily suffered a recorded loss. Removing such patients from the service comparison would make deferral look better than it is. We therefore retain the original endpoint and report pending obligations instead of adding a post hoc settlement period or noninferiority tolerance. The competitive-allocation results and training-only label-feasibility findings reinforce the need to distinguish loss prevention from completed service.

### *9.4. External validity and statistical limits*

All observations are synthetic. Patient deterioration, demand regimes, disruptions, resource costs, and transport assumptions have not been validated here against clinical outcomes or a deployed manufacturing network. Results cannot establish patient safety, real monetary savings, or performance under an organization's actual communication restrictions.

Five training blocks provide more information about training variability than additional evaluation worlds on the same trained pair, but they remain a small basis for generalization. The intervals are descriptive and unadjusted, the condition set is fixed, and the results have been examined during development. Passing a screen is not confirmation on an untouched external dataset. The exploratory three-block studies are still more limited and must not be used to strengthen the main estimate by pooling selected outcomes.

The structured bank also limits the method's action support. It does not search all admissible flows or future action sequences. Prediction futures and training tails introduce Monte Carlo uncertainty; matched stream identities are not proof of event-by-event equality. None of the saved diagnostics uniquely separates representation loss, stochastic continuation, and optimization error. Negative results constrain the tested recipes, not the entire class of graph models or patient-service objectives.

### *9.5. Generalization to simulation uncertainty*

The four primary regimes also occur during training; testing new random worlds from those regimes is not evidence of broad out-of-distribution generalization. The supplementary transfer and stale-advice tests in Appendix C cover particular perturbations and retain negative outcomes. They do not establish robustness to arbitrary changes in deterioration, manufacturing duration, supplier reliability, transport lead time, or the dependence between demand and disruptions.

A prospective uncertainty study would first separate parameter uncertainty within a calibrated model from distribution shift and model misspecification. It would distinguish frozen-policy transfer from retraining, prespecify application-supported ranges, keep information and computation matched, and report patient losses and unresolved service alongside cost. Changing many settings until a favorable graph result appears would not answer this question. No such study is launched or claimed as completed in this revision. The current evidence leaves this collaborator-prioritized research question open.

### *9.6. Relation to the earlier development record*

The repository retains earlier actor–critic DDPG and state-value TD/CEM implementations. DDPG [16], temporal-difference learning [15], value-augmented model predictive control [17], and CEM [18] describe different methodological components in that history. They are not alternative names for the paired residual estimator used here.

Earlier positive operational findings, inconclusive learning attribution, historical Flat comparisons, and adverse sensitivity results remain in their original records. They are not inserted into this panel or used as evidence for a new graph contribution. Likewise, changing the labels from losses and completions to losses and pending service supplies no new information by itself. A new objective weighting or input representation would be a new intervention requiring its own mechanism and prospective assessment.

The earlier three-block development results motivated the five-block replication; they did not follow an unexamined confirmatory selection process. The original five-block protocol was frozen before evaluation. A checkpoint cost-scale integer/float equivalence error stopped the first attempt before evaluation; the authorized recovery retained the original 120 world identities and completed only their missing evaluation, without selecting favorable prior test outcomes. This procedural recovery does not remove earlier model-selection or development-panel uncertainty.

Historical reporting counted five scoped Native campaign designs and fourteen configured learning arms, but those counts are not an exhaustive accounting of the project's learning-rate, architecture, target, and follow-up choices. An exact full-project selection count remains unresolved. Consequently, none of the reported intervals should be interpreted as correcting for the complete research search. Appendix B retains the historical Flat evidence instead of treating Self as a replacement that erases it.

## 10. Conclusion

The complete Graph/H8 controller improves on MDL2 in mean cost and patient outcomes on the five-block synthetic development panel. It also supplies a repeatable block-level cost increment over the same H8 planner, but not patient protection in every condition. The additional value of graph message passing over matched Self remains small and unstable in this strong-planner interface.

The separate matched-message study reveals a clearer graph cost effect: 4.843% lower mean cost and cost reductions in all three blocks, accompanied by 16.667 additional patient losses per world. This places graph representation at the center of a conditional design finding, not a claim of universal superiority. The later procurement result further shows that preserving each policy's own service does not establish equivalence between architectures. Together, the studies specify where the tested graph models add cost value and where that value fails to protect service. All findings remain synthetic development evidence, not clinical readiness or a proven service-preserving independent GCN gain.

## Data, code, and author review

The experiments use synthetic worlds. Code, frozen protocols, comparison files, and preservation records are maintained in the research repository. The accompanying collaborator-review package includes selected saved aggregate, block, condition, and paired-world comparison fields, their source hashes, and standalone figure-rebuilding code. This is a scoped evidence release, not a complete runnable release of simulator snapshots, training data, or checkpoints. Local source paths in the evidence map are identifiers, not proof that all underlying objects are available to the reader. No cloud-backup or clinical data-access claim is made. Authorship and affiliations are carried forward from the previous draft and require accountable-author confirmation. Contributions, funding, competing interests, ethics applicability, permissions, public availability, and venue-specific declarations remain unresolved before submission.

## Drafting tools and AI-assisted revision

OpenAI Codex assisted locally with source-to-method reconciliation, transcription of saved comparison fields, manuscript organization, and prose revision. The local scientific-writing and writing-review procedures were used as drafting aids. Scientific Agent Skills [27] is cited as a procedural resource, not as evidence for the manufacturing method or numerical results. This revision did not fit models, simulate outcomes, recompute comparisons, or validate clinical claims. Human review and approval remain pending; the authors must confirm the final AI-use disclosure and all declarations.

## Appendix A. Complete Block and Condition Summaries

The following tables transcribe saved comparison cells. G is Graph/H8, S is Self/H8, H is H8, and M is MDL2. Positive cost saving favors the candidate; positive loss delta is adverse and positive completion delta is favorable. $n_C$, $n_L$, and $n_D$ count worlds with higher cost, more losses, and fewer completions. Block denominators are 24; condition denominators are 30. No additional bootstrap or outcome aggregation was performed for this draft.

Table A.10: All block summaries for the six contrasts.

| Contrast | Block | Saving (%) | $\Delta L$ | $\Delta D$ | $n_C$ | $n_L$ | $n_D$ |
|---|---|---|---|---|---|---|---|
| G/M | 0 | 4.741 | -92.708 | +105.958 | 2 | 4 | 3 |
| G/M | 1 | 4.879 | -89.167 | +104.667 | 0 | 3 | 3 |
| G/M | 2 | 4.952 | -90.958 | +107.750 | 1 | 2 | 2 |
| G/M | 3 | 4.913 | -93.958 | +108.708 | 1 | 3 | 3 |
| G/M | 4 | 4.736 | -106.042 | +127.625 | 2 | 4 | 4 |
| G/H | 0 | 2.014 | +1.708 | +14.500 | 7 | 12 | 10 |
| G/H | 1 | 2.221 | -9.542 | +19.250 | 5 | 11 | 9 |
| G/H | 2 | 2.373 | -5.792 | +22.708 | 3 | 13 | 12 |
| G/H | 3 | 2.320 | -1.875 | +17.792 | 4 | 13 | 10 |
| G/H | 4 | 1.746 | +1.625 | +21.625 | 8 | 14 | 10 |
| G/S | 0 | 1.611 | -45.250 | +43.250 | 4 | 5 | 5 |
| G/S | 1 | 0.138 | +1.458 | -2.333 | 14 | 14 | 15 |
| G/S | 2 | 0.289 | -8.583 | +7.917 | 8 | 7 | 6 |
| G/S | 3 | -0.038 | +1.833 | -4.500 | 12 | 13 | 13 |
| G/S | 4 | 0.046 | +4.250 | +3.417 | 13 | 14 | 10 |
| S/H | 0 | 0.409 | +46.958 | -28.750 | 13 | 18 | 16 |
| S/H | 1 | 2.086 | -11.000 | +21.583 | 4 | 9 | 9 |
| S/H | 2 | 2.090 | +2.792 | +14.792 | 3 | 15 | 13 |
| S/H | 3 | 2.357 | -3.708 | +22.292 | 5 | 12 | 10 |
| S/H | 4 | 1.700 | -2.625 | +18.208 | 11 | 14 | 12 |
| S/M | 0 | 3.181 | -47.458 | +62.708 | 6 | 9 | 9 |
| S/M | 1 | 4.748 | -90.625 | +107.000 | 0 | 1 | 1 |
| S/M | 2 | 4.676 | -82.375 | +99.833 | 0 | 3 | 2 |
| S/M | 3 | 4.949 | -95.792 | +113.208 | 2 | 3 | 3 |
| S/M | 4 | 4.691 | -110.292 | +124.208 | 1 | 3 | 3 |
| H/M | 0 | 2.783 | -94.417 | +91.458 | 3 | 1 | 1 |
| H/M | 1 | 2.719 | -79.625 | +85.417 | 1 | 0 | 0 |
| H/M | 2 | 2.642 | -85.167 | +85.042 | 1 | 2 | 1 |
| H/M | 3 | 2.655 | -92.083 | +90.917 | 4 | 1 | 1 |
| H/M | 4 | 3.043 | -107.667 | +106.000 | 1 | 0 | 1 |

Table A.11: All condition summaries for the six contrasts. Conditions 0–3 are nominal history, abrupt regime shift, regional drift, and compound regional stress.

| Contrast | Cond. | Saving (%) | $\Delta L$ | $\Delta D$ | $n_C$ | $n_L$ | $n_D$ |
|---|---|---|---|---|---|---|---|
| G/M | 0 | 6.840 | -122.300 | +159.867 | 0 | 1 | 1 |
| G/M | 1 | 3.866 | -42.700 | +61.733 | 5 | 9 | 8 |
| G/M | 2 | 4.903 | -89.067 | +95.567 | 0 | 5 | 5 |
| G/M | 3 | 3.943 | -124.200 | +126.600 | 1 | 1 | 1 |
| G/H | 0 | 3.173 | -29.633 | +54.900 | 5 | 14 | 11 |
| G/H | 1 | 2.402 | +4.200 | +8.800 | 6 | 17 | 12 |
| G/H | 2 | 1.529 | +26.033 | -11.033 | 9 | 19 | 18 |
| G/H | 3 | 1.630 | -11.700 | +24.033 | 7 | 13 | 10 |
| G/S | 0 | 0.183 | -4.833 | +5.500 | 15 | 11 | 11 |
| G/S | 1 | 0.748 | -11.033 | +14.267 | 11 | 14 | 12 |
| G/S | 2 | 0.526 | -11.867 | +11.433 | 12 | 15 | 13 |
| G/S | 3 | 0.283 | -9.300 | +7.000 | 13 | 13 | 13 |
| S/H | 0 | 2.996 | -24.800 | +49.400 | 8 | 15 | 11 |
| S/H | 1 | 1.666 | +15.233 | -5.467 | 9 | 17 | 17 |
| S/H | 2 | 1.008 | +37.900 | -22.467 | 13 | 20 | 19 |
| S/H | 3 | 1.351 | -2.400 | +17.033 | 6 | 16 | 13 |
| S/M | 0 | 6.669 | -117.467 | +154.367 | 0 | 2 | 2 |
| S/M | 1 | 3.142 | -31.667 | +47.467 | 4 | 9 | 9 |
| S/M | 2 | 4.400 | -77.200 | +84.133 | 3 | 6 | 5 |
| S/M | 3 | 3.671 | -114.900 | +119.600 | 2 | 2 | 2 |
| H/M | 0 | 3.787 | -92.667 | +104.967 | 1 | 1 | 1 |
| H/M | 1 | 1.501 | -46.900 | +52.933 | 7 | 2 | 2 |
| H/M | 2 | 3.426 | -115.100 | +106.600 | 1 | 0 | 0 |
| H/M | 3 | 2.351 | -112.500 | +102.567 | 1 | 1 | 1 |

## Appendix B. Historical Architecture Evidence

Self removes hidden neighbor mixing while retaining the primary feature and planning interface. Historical Flat instead uses a globally informed MLP and answers a practical alternative-architecture question. Its stronger cross-site processing is not evidence of an invalid baseline. Neither control is weakened to secure a graph result.

Table B.12 preserves the earlier three-block, 72-world results. Graph and Flat were inherited frozen endpoints; Self was trained subsequently with the retained recipe. These are not the five independently trained pairs of the primary replication. A later 24-world comparison reused those same three pairs and yielded 1.053% Graph/Self saving and 22.333 fewer losses per world; it adds evaluation worlds, not three independent training replications. No five-block Flat supplement is reported as completed.

Table B.12: Explicitly historical comparisons on the earlier 72-world panel. Intervals are saved descriptive 95% cost intervals. Loss deltas are per-world means; parentheses count patient-harmed worlds out of 72.

| Candidate/reference | Saving (%) | Cost interval (%) | Extra losses |
|---|---|---|---|
| Graph/Self | 1.085 | [0.269, 2.178] | −25.750 (24) |
| Graph/Flat | 0.326 | [−0.514, 1.620] | +2.528 (46) |
| Graph/structured H8 | 2.754 | [1.862, 3.587] | −21.639 (29) |
| Graph/MDL2 | 4.557 | [3.687, 5.545] | −87.806 (13) |

On that panel Flat also improved on Self in both cost and mean losses, while the Graph/Flat interval crossed zero and Graph lost more patients than Flat on average. Thus the historical record does not establish superiority to a strong global MLP. Native-H8 from the earlier state-tail campaign and structured H8 from the Localv2 campaign also have different planners and candidate interfaces; their results cannot be pooled. These qualifications prevent the later Self ablation from concealing the earlier adverse architecture comparison.

## Appendix C. Perturbation and Transfer Boundaries

The following are completed, separate development studies, not new experiments for this revision. Their original world panels and saved decisions

remain distinct. None establishes broad simulation-uncertainty generalization.

Table C.13: Tested perturbations and retained negative boundaries.

| Study | Completed evidence and limitation |
|---|---|
| One withdrawn link | Five reused blocks, 40 paired seed tuples, 280 evaluations across original and changed networks. Target Graph/Self saving was 0.220% with interval [−0.092, 0.518]%; updating versus retaining old graph adjacency saved 0.031%, also with an interval crossing zero. Patient guards failed. The favorable 3.336% Graph/H8 result did not isolate GCN: Self/H8 saved 3.123%. |
| Multi-topology fitting | Three short-training blocks and 504 evaluations. On two held-out connection combinations, matched Graph/Self saving was 0.438% with interval [−0.018, 1.166]%, but Graph cost worsened by 0.105% relative to source-only short Graph. Self worsened by 0.245%. A wider architecture gap therefore did not establish absolute Graph improvement; fixed screens failed. |
| Cross-size preparation | Twelve fits and 216 source twenty-site evaluations completed. The source protection gate failed, so the planned 144 evaluations at 40/60 sites never ran. Larger-network performance remains unmeasured, not demonstrated inferior or superior. |
| Sample efficiency | Thirty checkpoints and 840 evaluations on 120 paired worlds. Graph trained on 64 worlds versus Self trained on 96 saved 0.266% with interval [−0.416, 1.004]%; no one-third training reduction was established. |

| Study | Completed evidence and limitation |
|---|---|
| Stale operational advice | Five reused blocks, 40 paired worlds, six roles, 240 evaluations. Candidates, H8 scores, and features were all delayed one week symmetrically. Stale Graph/Self saved 0.150% with interval [−0.473, 0.750]% and +7.500 losses. Stale Graph/H8 saved 9.026% with 281.425 fewer losses, but all controllers worsened relative to fresh advice. This is not measured network latency or evidence that GCN communicates faster. |

## Appendix D. Local Evidence and Version Boundaries

This appendix identifies the local records used for this working draft; they are not newly published external sources. Numeric rounding and fraction-to-percent display conversions do not constitute new estimates.

*E1: primary replication.* The result record is `results/native_graph_self_replication_20261017_eval_recovery1/payload/comparison.json`. Its `pairs`, block and condition fields support the six-contrast tables; its `table` fields supply endpoint and compute summaries. Completion, training reuse, and archive status come from `specs/2026-10-17-graph-self-replication/evaluation-recovery1/terminal-readout.json`. The recorded execution commit is `0ae59d25bdd3d46cd5778e2e0fd2c4c237767980`. Its archived evidence is not reverified by this manuscript edit.

*M1 and M2: method.* The five-block design is defined in `specs/2026-10-17-graph-self-replication/protocol-draft.md` and its effective proposal. Source implementations include `src/rl/native_candidate_advantage_targets.py`, `src/rl/native_candidate_advantage_learning.py`, `src/baselines/native_local_candidate.py`, `src/rl/native_structured_integration.py`, `src/models/native_local_candidate_value.py`, and `src/models/native_self_candidate_value.py`. The older three-block reference configuration supplies the inherited recipe, not the five-block evaluation denominator. Likewise, retained legacy TD configuration fields do not change the executed residual-target semantics.

*E2: separate exploratory studies.* The bounded-message result is recorded in `specs/2026-10-28-gcn-bounded-communication/result-readout.md`. Its communication contract and matched-model recipe are in the same directory's `protocol-draft.md`; historical prospective status there is superseded by the independent terminal record, not by manuscript wording. Procurement results and the archive failure are recorded in `specs/2026-10-30-gcn-procurement-preservation/reexecution1/terminal-readout.json`. They do not share the E1 panel. The separate completed archive-only closeout is recorded in `specs/2026-10-30-gcn-procurement-preservation/archive-completion1/terminal-readout.json`. Its new result root is `results/native_gcn_procurement_preservation_20261030_archive_completion1`. That receipt closes local preservation without changing the earlier failure record or any scientific estimate.

The procurement reexecution originally failed archive verification on its ledger-overhead limit. Its failed root and partial archive remain unchanged. The separately authorized closeout verified 2,299 members and 102,864,626 bytes; supervisor and archive child returned zero, with an external no-owned-process exit witness. It performed no new training, evaluation, forward calls, or comparisons. Local preservation does not establish cloud backup or collaborator access.

*E3: negative development record.* The internal index `reports/2026-10-30-gcn-next-stage/evidence-map.md` links completed precision, decision-loss, auxiliary, and conditional-query readouts. The intervention record is `reports/2026-10-30-cost-service-feasibility/recorded-interventions.md`; the training-only feasibility decision is `reports/2026-10-30-service-signal-feasibility/readout.md`. The earlier DDPG and TD/CEM manuscripts, historical Flat comparisons, and four-site support-hour material remain separate and unchanged. The prior `main.tex` is preserved in Git history. None of those files should be read as the current Localv2 specification or pooled with E1.

*E4: supplementary export.* The latest conditional-recipient closeout is `specs/2026-10-30-gcn-conditional-recipient-allocation/assessment-closeout1/terminal-readout.json`; the follow-up decision inventory is `reports/2026-10-31-joint-request-reuse-inventory/decision.json`. The manuscript's `data/source-manifest.json` binds selected comparison exports and copied supplementary readouts to their original bytes and

hashes. `REVIEW-20261031.md` explains which evidence is included and the limits of standalone reproduction. The related-work source register separately records assistant access limits and pending accountable-human verification; no such human approval is inferred from successful compilation.

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